

Venkatesh Kelam

☎ (669)-236-9712 ✉ venkateshkelam01@gmail.com [in linkedin.com/in/venkatesh0101](https://www.linkedin.com/in/venkatesh0101) github.com/venkateshkelam01

Summary

Graduate student in Information Systems and AWS Certified Solutions Architect with experience developing and automating distributed cloud systems. Skilled in **Java, Python, Go, and C++** with expertise in **AWS, Terraform, and Kubernetes**. Passionate about cloud storage, container orchestration, and optimizing data systems for reliability and scale.

Education

Master of Science in Information Systems | Northeastern University, San Jose, CA Aug 2023 – Present

B.Tech in Information Technology | Anna University, Chennai Apr 2023

Technical Skills & Certification

Languages: Java, Python, JavaScript, TypeScript, C++, Bash, SQL

Frameworks: Spring Boot, React.js, FastAPI, LangChain, TensorFlow, PyTorch

Cloud & Tools: AWS (EC2, S3, CloudWatch, DynamoDB), Docker, Kubernetes, Jenkins, Terraform, CloudFormation

Databases: MongoDB, PostgreSQL, Redis, MySQL

DevOps & Systems: CI/CD pipelines, Linux/Unix internals, Cloud Storage, IaC (Terraform/CloudFormation)

Certification: AWS Certified Solutions Architect – Associate

Work Experience

Software Engineer Intern

Feb 2025 – May 2025

Ipser Labs, Remote

- Developed and deployed **Dhorobha**, a scalable cloud-native real estate platform hosted on **AWS EC2** and **S3**.
- Automated infrastructure provisioning using **Terraform** for consistent, repeatable cloud deployments.
- Configured CI/CD pipelines using **Jenkins** to streamline testing and deployment for backend microservices.
- Designed and optimized **RESTful APIs** in **Spring Boot**, achieving a 200ms average response time under load.
- Integrated **MongoDB** and **PostgreSQL** for hybrid data storage, improving query speed and scalability.
- Enhanced system reliability through **CloudWatch** metrics and log-based monitoring dashboards.

Front-End Developer Intern

Apr 2022 – Apr 2023

Uniq Solutions, Chennai, India

- Built interactive healthcare dashboards using **React.js** with APIs powered by **AWS Lambda** and **MongoDB Atlas**.
- Resolved Tier-2 dashboard incidents by triaging issues, reproducing defects, and deploying hotfixes within defined SLAs.
- Collaborated with backend teams to automate build and deployment workflows using **Jenkins** pipelines.
- Assisted in migrating on-prem systems to cloud-hosted environments, improving uptime and scalability.
- Participated in Agile sprints and code reviews, contributing to a 30% reduction in release cycle time.

Projects

RideShare Cloud Platform | Python, Node.js, AWS, Kubernetes, Terraform, PostgreSQL

Feb 2025 – Present

- Implemented a distributed microservice architecture in **Python** for a cloud-native storage platform, leveraging **Kubernetes StatefulSets** and **Persistent Volume Claims (PVCs)** to manage dynamic data volumes; optimized I/O throughput and latency using **gRPC streaming** and connection pooling across nodes.
- Automated cross-cloud provisioning and deployments using **Terraform** and **Jenkins CI/CD**, integrating health checks, rollback triggers, and CloudWatch-based anomaly alerts to ensure >99.9% uptime.
- Configured **AWS CloudWatch** metrics and distributed tracing pipelines for real-time service reliability & fault detection.

EasyEmail — AI Email Assistant | Python, FastAPI, LangChain, PyTorch, Docker

Dec 2024

- Developed **storage analytics and monitoring microservices** in **Python** to track read/write latency, throughput, and cache hit ratios; integrated **Prometheus + Grafana** dashboards with log-based alerts for proactive SLA tracking.
- Built an AI email assistant using **FastAPI**, **LangChain**, and **PyTorch** for intelligent text parsing & semantic search.
- Containerized and deployed services using **Docker**, automated via **Jenkins** and event triggers on **AWS Lambda**.

Healixity — Cloud Health Assistant | Go, Next.js, TensorFlow, DynamoDB, Pinecone

Feb 2025

- Automated multi-cloud infrastructure provisioning (**AWS + Azure**) using reusable **Terraform modules** with integrated **Jenkins CI/CD** pipelines for zero-downtime deployments.
- Developed data retrieval APIs in **Go** using **AWS DynamoDB** and **Pinecone**, improving semantic search latency.
- Implemented infrastructure observability with **CloudWatch** metrics and performance dashboards to monitor resource utilization and latency trends.